

Command Reference

Navigation

Command	Usage	Common Flags	Example
cd	change directory	-, ~	cd /var/log
pushd	push dir onto stack, cd into it	+N (rotate to Nth)	pushd /etc/nginx
popd	pop dir off stack, cd back	+N	popd
pwd	print working directory	-P (resolve symlinks)	pwd -P
ls	list directory contents	-l, -a, -h, -t	ls -lah

System Info

Command	Usage	Common Flags	Example
uname	print system info	-a (all), -r (kernel release), -m (arch)	uname -a
df	disk space usage	-h (human-readable), -T (show fs type)	df -h
uptime	time running + load average	-p (pretty), -s (since boot)	uptime
free	memory usage	-h (human-readable), -s (repeat interval)	free -h
lscpu	CPU architecture details	n/a	lscpu
lsblk	list block devices/partitions	-f (show filesystems)	lsblk -f
lsmem	show memory range/block info	-a (all ranges), -s (summary)	lsmem
lshw	detailed hardware info (CPU, memory, disks, network)	-short (summary), -C (class filter)	lshw -short
cat /etc/os-release	show distro name/version	n/a	cat /etc/os-release

Debugging

Command	Usage	Common Flags	Example
dmesg	print kernel ring buffer (boot/hardware/driver)	-T (human timestamps), -w (follow)	dmesg -T tail

	messages)	live), -l err (filter level)	-20
less	page through file/output content	-N (line numbers), -S (no line wrap), /pattern (search)	dmesg -T less
grep	search text by pattern	-i (case- insensitive), -r (recursive), -n (line numbers), -v (invert match)	dmesg -T grep -i error
tail	show end of file/output	-f (follow live), -n (number of lines)	tail -f /var/log/syslog
head	show start of file/output	-n (number of lines)	head -n 20 file.log
udevadm	query/control udev device manager	info (query device), monitor (live events), trigger (re- run rules)	udevadm monitor
lscpu	CPU architecture details	n/a	lscpu
lsblk	list block devices/partitions	-f (show filesystems)	lsblk -f
lspci	list PCI devices	-v (verbose), -k (show kernel driver)	lspci -k
lsmem	show memory range/block info	-a (all ranges), -s (summary)	lsmem
lshw	detailed hardware info (CPU, memory, disks, network)	-short (summary), -C (class filter)	lshw -short

File Operations

Command	Usage	Common Flags	Example
mkdir	create directory	-p (create parents)	mkdir -p a/b/c
touch	create empty file / update timestamp	-a, -m	touch file.txt
rm	remove file/directory	-r (recursive), -f (force), -i (confirm)	rm -rf dir/
cp	copy	-r (recursive), -p (preserve attrs), -i (confirm), -- parents	cp -r src/ dst/

	file/directory	(recreate source path structure at destination)	
mv	move/rename file/directory	-i (confirm), -n (no overwrite)	mv file.txt newname.txt
chmod	change permissions	-R (recursive), +x, 755	chmod -R 755 dir/
chown	change owner/group	-R (recursive)	chown -R user:group dir/
chgrp	change group only	-R (recursive)	chgrp android test-file
cat	print/concat file contents	-n (line numbers), -A (show hidden chars)	cat -n file.txt

ACL

Command	Usage	Common Flags	Example
setfacl -m	set/modify an ACL entry	u: (user), g: (group)	setfacl -m u:bob:rw file.txt
setfacl -x	remove a specific ACL entry	u: (user), g: (group)	setfacl -x u:bob file.txt
setfacl -b	remove all ACL entries	-b (blank/remove all)	setfacl -b file.txt
getfacl	view ACL entries on a file/directory	n/a	getfacl file.txt

Command Lookup

Command	Usage	Common Flags	Example
man	show manual page	-k (keyword search)	man ls
apropos	search man page descriptions by keyword	-e (exact match)	apropos copy
type	show how name resolves (alias/builtin/file)	-a (all matches)	type cd
which	show path to executable	-a (all matches)	which python3
whatis	one-line description of command	-r (regex search)	whatis ls
help	bash builtin help	-d (short desc), -m (man-style)	help cd
info	detailed GNU manual	-a (all matches)	info coreutils
--help / -h	flag (not standalone command) —	n/a	ls --help, tar

most commands
print their own
usage summary

-h

Output / Text

Command	Usage	Common Flags	Example
echo	print text to stdout	-n (no newline), -e (interpret escapes)	echo -e "line1\nline2"
less	page through file/output content	-N (line numbers), -S (no line wrap), /pattern (search)	less file.txt
grep	search text by pattern	-i (case-insensitive), -r (recursive), -n (line numbers), -v (invert match)	grep -rn "TODO" src/
tail	show end of file/output	-f (follow live), -n (number of lines)	tail -n 20 file.log
head	show start of file/output	-n (number of lines)	head -n 20 file.log

Searching Files & Patterns

Command	Usage	Common Flags	Example
locate	fast file search using pre-built database	n/a	locate City.txt
updatedb	refresh the locate database	n/a	sudo updatedb
find	search filesystem by name, date, size, etc (no database)	-name, -mtime, -size	find /home/michael -name City.txt
grep	search text by pattern	-i (case-insensitive), -R (recursive), -v (invert match), -w (whole word), -A (lines after), -B (lines before)	grep -R "third line" /home/michael

Compression & Archival

Command	Usage	Common Flags	Example
du	show file/directory size	-s (summary), -k (KB), -h (human-readable)	du -sh test.img

tar -c	create an archive	-c (create), -f (filename), -z (gzip compress)	tar -cf test.tar file1 file2
tar -t	list contents of an archive	-t (list), -f (filename)	tar -tf test.tar
tar -x	extract an archive	-x (extract), -f (filename), -z (gzip decompress)	tar -xf test.tar
bzip2	compress a file (.bz2)	n/a	bzip2 test.img
bunzip2	decompress a .bz2 file	n/a	bunzip2 test.img.bz2
bzcat	view .bz2 contents without decompressing	n/a	bzcat test.img.bz2
gzip	compress a file (.gz)	n/a	gzip test1.img
gunzip	decompress a .gz file	n/a	gunzip test1.img.gz
zcat	view .gz contents without decompressing	n/a	zcat test1.img.gz
xz	compress a file (.xz)	n/a	xz test2.img
unxz	decompress a .xz file	n/a	unxz test2.img.xz
xzcat	view .xz contents without decompressing	n/a	xzcat test2.img.xz

Disk Partitioning

Command	Usage	Common Flags	Example
fdisk -l	list partition tables (MBR/legacy)	-l (list)	sudo fdisk -l /dev/sda
gdisk	create/edit GPT partitions (interactive)	n (new), d (delete), p (print table), w (write & exit)	gdisk /dev/sdb

File Systems & Mounting

Command	Usage	Common Flags	Example
mkfs.ext4	format a partition with ext4 filesystem	n/a	mkfs.ext4 /dev/sdb1
mount	mount a filesystem to a directory	n/a	mount /dev/sdb1 /mnt/ext4
blkid	print UUID of block devices	n/a	blkid /dev/sdb1

NFS

Command	Usage	Common Flags	Example
exportfs -a	export all shares defined in /etc/exports	-a (all)	exportfs -a
exportfs -o	export a directory to a specific client	-o (options/client)	exportfs -o 10.61.35.201:/software/repos
mount	mount an NFS share from a remote server	server:/path local_mount	mount 10.61.112.101:/software/repos /mnt/software/repos

LVM

Command	Usage	Common Flags	Example
pvccreate	create a physical volume from a disk/partition	n/a	pvccreate /dev/sdb
pvddisplay	show physical volume details	n/a	pvddisplay
vgcreate	create a volume group from physical volumes	n/a	vgcreate my_vg /dev/sdb
vgdisplay	show volume group details	n/a	vgdisplay
vgs	summary of volume groups (size, free space)	n/a	vgs
lvcreate	create a logical volume within a volume group	-L (size), -n (name)	lvcreate -L 1G -n vol1 my_vg
lvs	list logical volumes	n/a	lvs
lvresize	resize a logical volume	-L +Xg (grow by size)	lvresize -L +1G /dev/my_vg/vol1
resize2fs	resize filesystem to match resized LV	n/a	resize2fs /dev/my_vg/vol1

IO Redirection

Technique	Usage	Common Flags	Example
>	redirect STDOUT to a file (overwrite) append	n/a	echo \$SHELL > shell.txt

>>	STDOUT to a file	n/a	echo \$SHELL >> shell.txt
2>	redirect STDERR to a file (overwrite)	n/a	cat missing_file 2> error.txt
2>>	append STDERR to a file	n/a	cat missing_file 2>> error.txt
/dev/null	discard output (used as redirect target)	n/a	command 2> /dev/null
	pipe STDOUT of one command into another	n/a	grep Hello sample.txt less
tee	duplicate output to file and screen	-a (append instead of overwrite)	echo \$SHELL tee shell.txt

User Management

Command	Usage	Common Flags	Example
adduser	create new user (interactive)	--home, --shell, --disabled-password	adduser bob
groupadd	create new group	-g (set GID)	groupadd devs
useradd	create user (non-interactive)	-u (UID), -g (primary GID), -d (home dir), -s (shell), -c (comment), -e (expiry date), -G (secondary groups)	useradd -u 1009 -g 1009 -d /home/robert -s /bin/bash -c "Mercury Project member" bob
usermod	modify existing user	-aG (add to group), -s (change shell), -L (lock)	usermod -aG sudo bob
userdel	delete a user	-r (remove home dir)	userdel -r bob
groupdel	delete a group	n/a	groupdel devs
passwd	set/change password	-l (lock), -e (expire)	passwd bob
su	switch user	- (login shell)	su - bob
su -c	run a single command as another user	-c (command)	su -c "whoami"
sudo	run command as another user	-u (user), -s (shell)	sudo -u bob whoami
id	show user/group IDs	-u, -g, -Gn	id -u bob
whoami	show current user	n/a	whoami
groups	show group memberships	n/a	groups bob

Networking

Command	Usage	Common Flags	Example
ping	test connectivity to a host	-c (count), -i (interval)	ping db
nslookup	query DNS server for a hostname	n/a	nslookup www.google.com
dig	detailed DNS query and record info	+short (brief output)	dig www.google.com
ip link	list/modify host interfaces	n/a	ip link
ip addr	show IP addresses assigned to interfaces	n/a	ip addr
ip addr add	assign an IP address to an interface	n/a	ip addr add 192.168.1.10/24 dev eth0
route	view the kernel routing table	n/a	route
ip route	view the routing table (iproute2)	n/a	ip route
ip route add	add an entry to the routing table	default via (default gateway)	ip route add default via 192.168.2.1
traceroute	trace the network route/hops to a host	n/a	traceroute caleston-repo-01
netstat	show network connections/listening ports	-a (all), -n (numeric), -t (tcp)	netstat -an grep 80
ip link set	bring an interface up/down	up, down	ip link set dev enp1s0f1 up

Firewall

Command	Usage	Common Flags	Example
iptables -L	list current rules by chain (INPUT, FORWARD, OUTPUT)	-L (list)	sudo iptables -L
iptables -A	append a rule to a chain	-p (protocol), -s (source), -d (dest), --dport (port), -j (target: ACCEPT/DROP)	iptables -A INPUT -p tcp -s 172.16.238.187 --dport 22 -j ACCEPT
iptables -I	insert a rule at the top of a chain (takes precedence)	same as -A	iptables -I OUTPUT -p tcp -d 172.16.238.100 --dport 443 -j ACCEPT
iptables -D	delete a rule by chain + position	n/a	iptables -D OUTPUT 5

-A / -I expect a chain name right after them:
INPUT -> incoming traffic

OUTPUT -> outgoing traffic
FORWARD -> routed traffic (through the host, not to/from it)

-p expects a protocol:
tcp, udp, icmp, all

-j expects a target (what to do with matching packets):
ACCEPT -> allow
DROP -> silently discard
REJECT -> discard + send error back

e.g. iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables -I OUTPUT -p tcp -d 1.2.3.4 --dport 443 -j DROP

Note: a “chain” (INPUT, OUTPUT, FORWARD) is just a list of rules — packets are checked against each rule in order until one matches. Make sure to actually use -A (append) or -I (insert) when adding a rule; without one of these, the command won’t add anything to the chain.

Date / Time

Command	Usage	Common Flags	Example
timedatectl	show current date/time/timezone status	n/a	timedatectl
timedatectl list-timezones	list all valid timezones	n/a	timedatectl list-timezones
timedatectl set-timezone	set system timezone	n/a	timedatectl set-timezone Asia/Manila

SSH / SCP

Command	Usage	Common Flags	Example
ssh	remote login/command execution	-l (specify user), user@host syntax	ssh bob@devapp01
ssh-keygen	generate SSH key pair	-t (key type, e.g. rsa)	ssh-keygen -t rsa
ssh-copy-id	copy public key to remote server’s authorized_keys	n/a	ssh-copy-id bob@devapp01
scp	securely copy files/dirs over SSH	-r (recursive), -p (preserve perms/ownership)	scp file.tar.gz devapp01:/home/bob

Note: to disable root login over SSH, edit PermitRootLogin in /etc/ssh/sshd_config:

PermitRootLogin no

Options: yes (allow), no (disable), prohibit-password (key-only, no password), forced-commands-only (root can only run a forced command). Restart sshd after changing: sudo systemctl restart sshd

Systemd Services

Command	Usage	Common Flags	Example
systemctl start	start a service	n/a	sudo systemctl start project-mercury.service
systemctl stop	stop a service	n/a	sudo systemctl stop project-mercury.service
systemctl restart	stop then start a service	n/a	systemctl restart docker
systemctl reload	reload service config without interrupting it	n/a	systemctl reload docker
systemctl status	show service status	n/a	sudo systemctl status project-mercury.service
systemctl enable	enable service to start on boot	n/a	sudo systemctl enable project-mercury.service
systemctl disable	prevent service from starting on boot	n/a	systemctl disable docker
systemctl daemon-reload	reload systemd after editing a unit file	n/a	sudo systemctl daemon-reload
systemctl edit --full	edit a unit file, applies immediately on save	--full (edit full file, not override)	systemctl edit project-mercury.service --full
systemctl list-units --all	list all units, loaded or not	--all (include inactive)	systemctl list-units --all
journalctl -u	view logs for a specific unit	-u (unit name)	journalctl -u docker.service

Note: service unit files live in /etc/systemd/system. To create a custom service, just add a new file there ending in .service.

Example: sudo vim /etc/systemd/system/my-app.service

Service Unit File Skeleton

```

[Unit]
Description=Short description of the service
Documentation=http://example.com/docs
After=postgresql.service

[Service]
ExecStart=/bin/bash /usr/bin/my-script.sh
User=service_account
Restart=on-failure
RestartSec=10

[Install]
WantedBy=graphical.target

```

Targets (Runlevels)

Command	Usage	Common Flags	Example
systemctl get-default	show current default target	n/a	systemctl get-default
systemctl set-	set default		systemctl set-

default	target	n/a	default multi-user.target
systemctl list-units --type=target	list active targets	n/a	systemctl list-units --type=target
systemctl list-unit-files --type=target	list all available targets	n/a	systemctl list-unit-files --type=target

System Config Locations

Path	Usage
/etc/passwd	user account info
/etc/shadow	hashed passwords
/etc/group	group definitions
/etc/sudoers	sudo permissions (edit via visudo)
/etc/hosts	static hostname-to-IP mapping
/etc/fstab	filesystem mount config
/etc/exports	NFS shared directories and allowed clients
/etc/ssh/sshd_config	SSH server config
/etc/resolv.conf	DNS resolver config
/etc/systemd/system/	custom systemd unit files
/etc/crontab	system-wide cron jobs
/etc/nsswitch.conf	name resolution order (hosts, DNS, etc.)
/sbin/init	first process started by kernel at boot (often symlink to systemd)

Top-Level Directories

Path	Usage
/bin	essential user binaries
/sbin	essential system binaries (admin tools)
/etc	system-wide config files
/home	user home directories
/root	root user's home directory
/var	variable data (logs, spool, cache)
/tmp	temporary files (cleared on reboot)
/usr	user programs, libraries, docs
/opt	optional/third-party software
/mnt	temporary mount points
/media	removable media mount points
/dev	device files
/proc	live kernel/process info (virtual)
/sys	kernel device/driver info (virtual)
/boot	boot loader files, kernel images
/lib	shared libraries

File Type Symbols

Symbol	Meaning
--------	---------

-	regular file
d	directory
l	symbolic link
c	character device
b	block device
s	socket
p	named pipe (FIFO)

Crontab Format

Command	Usage	Common Flags	Example
<code>crontab -e</code>	edit current user's crontab	-u (specify user; needs root)	<code>crontab -e</code>
<code>crontab -l</code>	list current user's cron jobs	-u (specify user; needs root)	<code>crontab -l</code>

Note: using `sudo crontab -e` edits root's crontab, not your own. Omit `sudo` to edit your own user's crontab.

Note: access to `crontab` can be restricted via `/etc/cron.allow` and `/etc/cron.deny`. - If `cron.allow` exists, only users listed in it can use `crontab` (everyone else is blocked). - If `cron.allow` doesn't exist, users listed in `cron.deny` are blocked; everyone else is allowed. - If neither file exists, only root can use `crontab` (varies by distro).

```
* * * * * command_to_run
| | | | |
| | | | +---- day of week (0-6, Sun=0)
| | | +----- month (1-12)
| | +----- day of month (1-31)
| +----- hour (0-23)
+----- minute (0-59)
```

Permissions (chmod Octal Values)

```
r (read)    = 4
w (write)   = 2
x (execute) = 1
no perms    = 0
```

```
rwX = 4+2+1 = 7
r-X = 4+0+1 = 5
-wX = 0+2+1 = 3
```

```
chmod [owner][group][others] file
e.g. chmod 750 file → owner: rwx, group: r-x, others: ---
```

Process Limits

Command	Usage	Common Flags	Example
<code>ulimit -a</code>	show all current limits	-a (all)	<code>ulimit -a</code>
<code>ulimit -n</code>	set max open file descriptors set max	n/a	<code>ulimit -n 4096</code>

ulimit -u	processes for user	n/a	ulimit -u 2048
ulimit -S / -H	set/show soft vs hard limit	-S (soft), -H (hard)	ulimit -Hn

Note: ulimit changes are temporary (current shell session only).
For persistent limits, edit /etc/security/limits.conf:

```
username soft nproc 2048
username hard nproc 4096
```

Other common resource types (same field, swap in for nproc):

```
nofile -> max open file descriptors
fsize  -> max file size (KB)
cpu     -> max CPU time (minutes)
as      -> max address space / virtual memory (KB)
memlock -> max locked-in-memory address space
stack   -> max stack size (KB)
nice    -> max nice priority allowed
rtprio  -> max real-time priority
maxlogins -> max simultaneous logins for a user
```

Package Management

APT

Command	Usage	Common Flags	Example
apt update	refresh package repository information	n/a	apt update
apt upgrade	upgrade installed packages to latest version	n/a	apt upgrade
apt install	install a package, resolves dependencies automatically	n/a	apt install telnet
apt remove	remove an installed package	n/a	apt remove telnet
apt search	search for a package within the repository	n/a	apt search telnet
apt edit-sources	edit repository sources	n/a	apt edit-sources

DPKG

Command	Usage	Common Flags	Example
dpkg -i	install or update a package	-i (install)	dpkg -i telnet.deb
dpkg -r	remove a package	-r (remove)	dpkg -r telnet
dpkg -l	list installed packages with version/description check status of an	-l (list)	dpkg -l telnet

`dpkg -s` installed package -s (status) `dpkg -s telnet`

YUM

Command	Usage	Common Flags	Example
<code>yum repolist</code>	list configured repositories	n/a	<code>yum repolist</code>
<code>yum provides</code>	find which package provides a file/command	n/a	<code>yum provides scp</code>
<code>yum remove</code>	remove a package	n/a	<code>yum remove httpd</code>
<code>yum update <package></code>	update a specific package	n/a	<code>yum update telnet</code>
<code>yum update</code>	update all packages	n/a	<code>yum update</code>

RPM

Command	Usage	Common Flags	Example
<code>rpm -ivh</code>	install a package	-i (install), -v (verbose), -h (progress)	<code>rpm -ivh package.rpm</code>
<code>rpm -e</code>	uninstall a package	-e (erase)	<code>rpm -e package</code>
<code>rpm -Uvh</code>	upgrade an existing package	-U (upgrade), -v (verbose), -h (progress)	<code>rpm -Uvh package.rpm</code>
<code>rpm -q</code>	query installed package details	n/a	<code>rpm -q package</code>
<code>rpm -Vf</code>	verify installed package against original metadata	-V (verify), -f (by file path)	<code>rpm -Vf <path to file></code>

Notes

- When making a script executable (`chmod +x`), make sure it also has read permission (`r`). Without read access, the kernel can't load the file's contents to run it, so it will still fail to execute even with the execute bit set.
- Use `$(command)` to substitute the output of one command as an argument to another. e.g. `chsh -s $(which nologin) bob`.
- `find` has an `-exec` option that runs a command on each matched result. Use `{}` as a placeholder for the found item and end with `\;`. e.g. `find . -name "*.log" -exec rm {} \;`.
- For `cp`, don't confuse `-p` and `--parents`: `-p` preserves file attributes (timestamps, permissions, ownership); `--parents` recreates the full source path structure at the destination. e.g. `cp --parents dir1/dir2/file.txt /backup/` creates `/backup/dir1/dir2/file.txt`.